



SUMMER INTERNSHIP

Java Full Stack Developer Internship

Name: Ayush

Roll Number CS-2341826

Program & Semester B.Tech CSE - 7th Sem

Organization Name EduSkills (AICTE)

Domain Enterprise Java

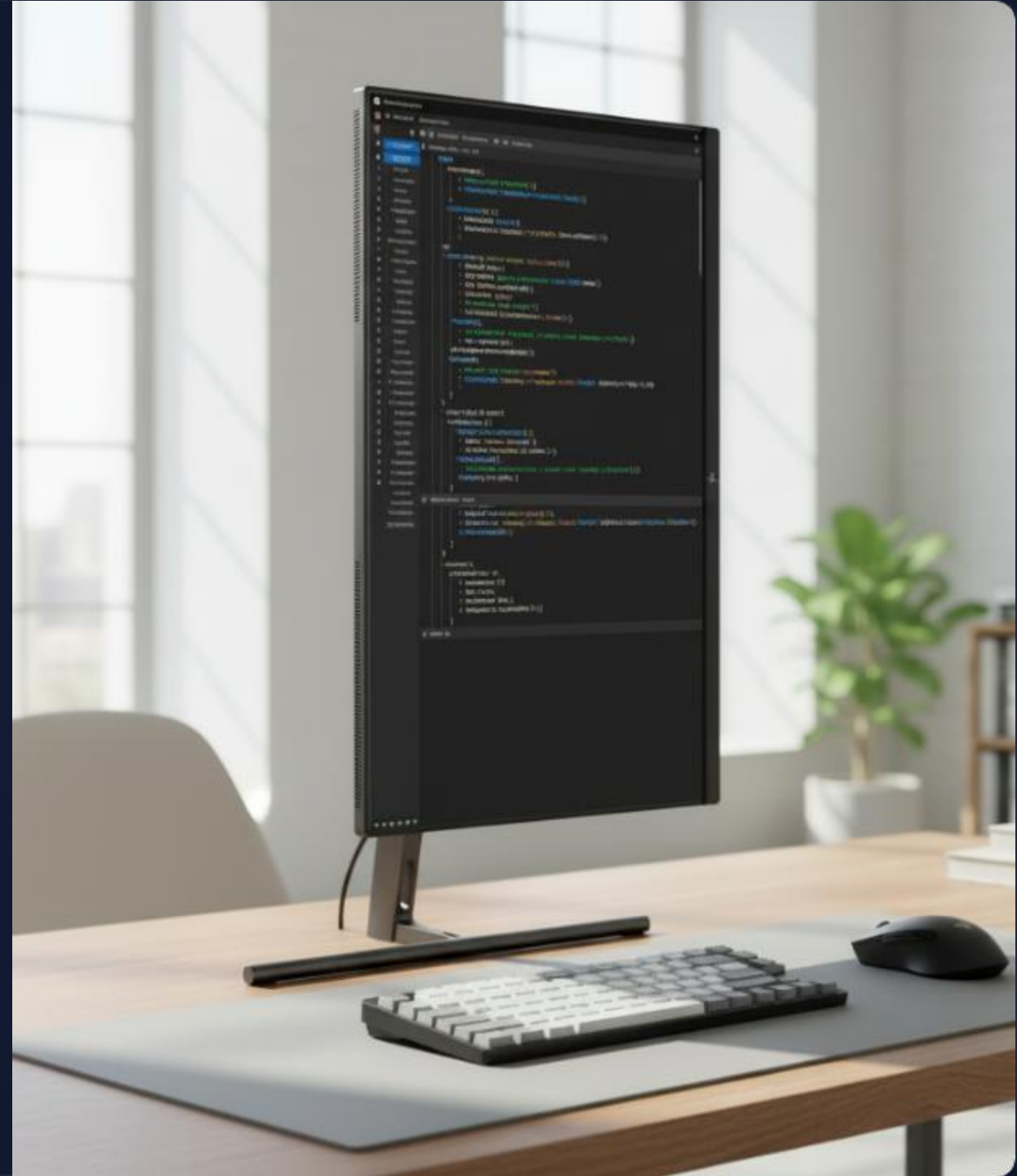
Academic Year 2026-27

EduSkills Profile

Nation Building Through Skills

EduSkills, a non-profit organization supported by AICTE, seeks to bridge the critical gap between academic theory and the demands of corporate technology.

They are an ISO 9001:2015 certified organization and tie up with global technology giants to offer structured virtual internships that are relevant to the industry.



Three Pillars of Full Stack Development



Presentation

Architecting responsive, mobile-first frontend interfaces utilizing HTML5, CSS3, Bootstrap, and JavaScript to manage user interactions dynamically.



Application

Engineering enterprise-grade backend REST APIs by leveraging Core Java, Servlets, and the powerful Spring Boot framework for business logic.



Data Access

Implementing secure, optimized database operations utilizing MySQL alongside the Hibernate Object-Relational Mapping (ORM) architecture.

Phase 1: Frontend Engineering

Weeks 1-3: Constructing the Presentation Layer and User Interfaces

Foundations: HTML5 & CSS3

Semantic HTML5

The internship began by laying the foundation for basic UI design. This included the removal of common 'div' tags and the use of the semantic HTML5 elements (header, main, section, footer).

The structure will provide top-tier accessibility with screen readers and greatly enhance the Search Engine Optimization (SEO) of web applications.

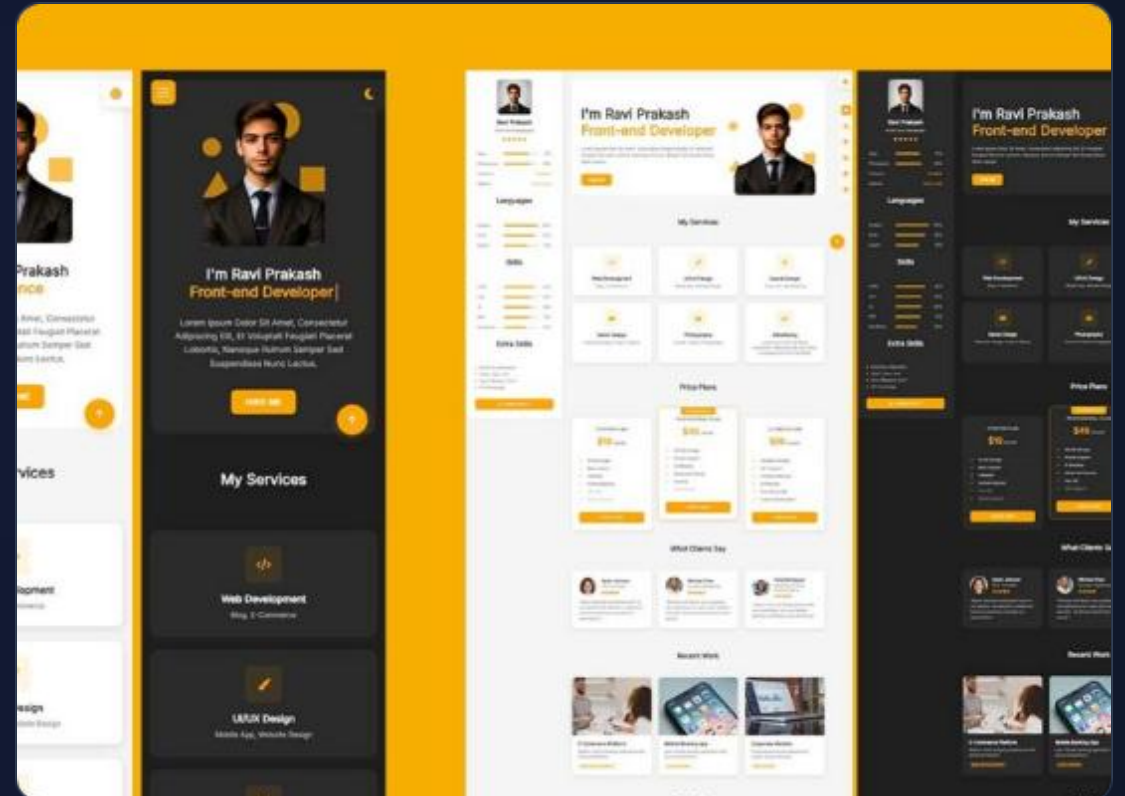
Advanced CSS3

Advanced CSS mechanics were explored to control complex layouts without relying on external libraries initially.

There was lots of time spent learning more about the CSS Box Model, Flexbox for working with one dimensional layouts, and CSS Grid for positioning more complex 2 dimensional components.

Bootstrap & Responsive Design

- ▶ **Rapid Prototyping:** Rewrote all the UI with Bootstrap framework to speed up the UI development and to ensure consistency.
- ▶ **Mobile-First Architecture:** Learned the responsive 12 column responsive grid system, enabling Web applications to adjust to viewports of mobile, tablet, and desktop.
- ▶ **Bootstrap utility classes for pre-built components:** Created interactive components such as collapsing navbars, modals, and carousels with components from the Bootstrap utility classes.
- ▶ **Cross Browser Compatibility:** Used normalized stylesheets to create UI designs that were consistent across Chrome, Firefox and Safari.



Dynamic Scripting: JavaScript & jQuery



Core JS Mechanics: Mastered ES6+ syntax including block scoping (let/const), arrow functions, and higher-order array methods (map, filter, reduce).



DOM Manipulation: Developed ability to detect user events (clicks, keypresses) and dynamically update HTML text and styles on the fly without refreshing the page.



Client-Side Validation: Implemented logic to prevent default form submissions so that user input is validated securely before hitting the backend server.



jQuery Integration: Used the jQuery library to reduce the amount of DOM traversal code and make asynchronous operations (AJAX) easier.

Phase 2: Enterprise Backend

Weeks 4-6: Business Logic and Application Layers

Core Java Fundamentals

Object-Oriented Paradigms

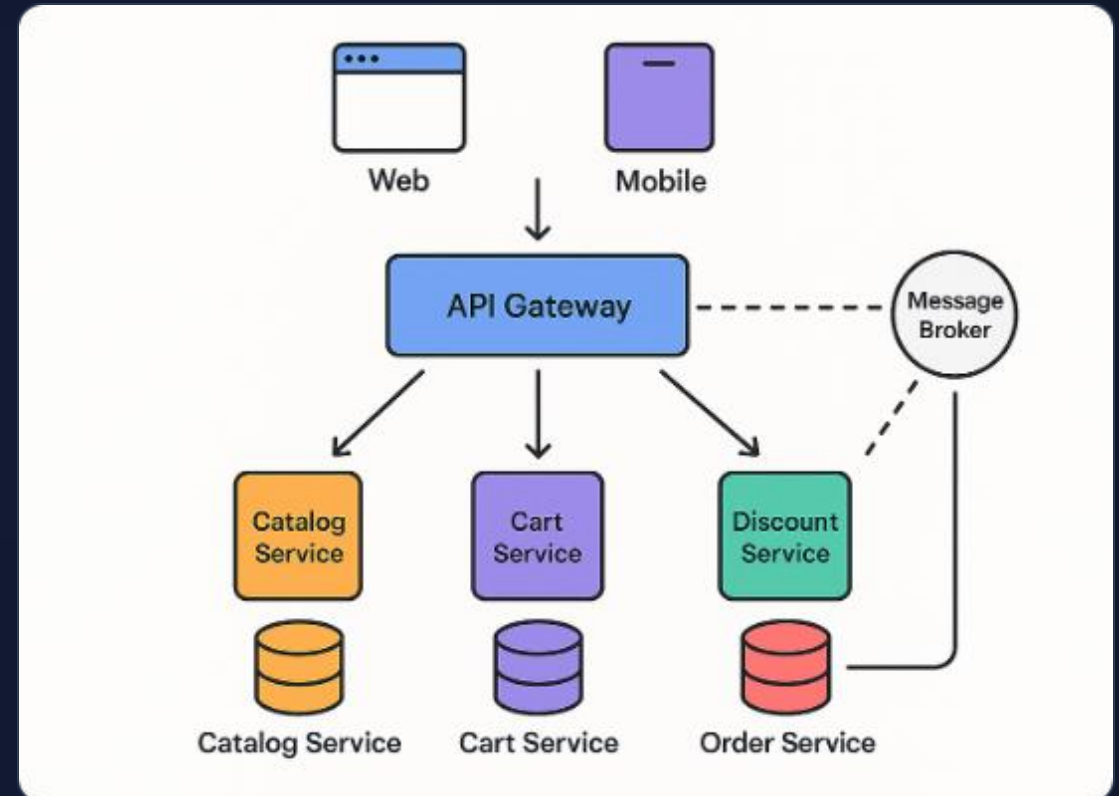
The backend phase began with a rigorous application of OOP. We moved beyond basic syntax to understand how enterprise software utilizes Encapsulation to protect data integrity, Inheritance for code reusability, and Polymorphism for dynamic method resolution.

Data Structures & Memory

Mastered the Java Collections Framework (Lists, Maps, Sets) to ensure highly efficient data storage and retrieval in memory. Coupled this with robust Exception Handling (try-catch-finally) to build fault-tolerant backend services that don't crash unexpectedly.

Advanced Java (Servlets & JDBC)

- ▶ **Web Containers:** Deployed applications to Apache Tomcat, the web server.
- ▶ **Servlet Lifecycle:** Handled raw HTTP GET and POST requests by overriding specific Servlet methods (init, service, destroy).
- ▶ **JDBC Integration:** Set up secure connections from the Java backend to relational databases with DriverManagers.
- ▶ **Security:** Used PreparedStatement to safely execute SQL statements, resulting in no risk of SQL injection attacks.



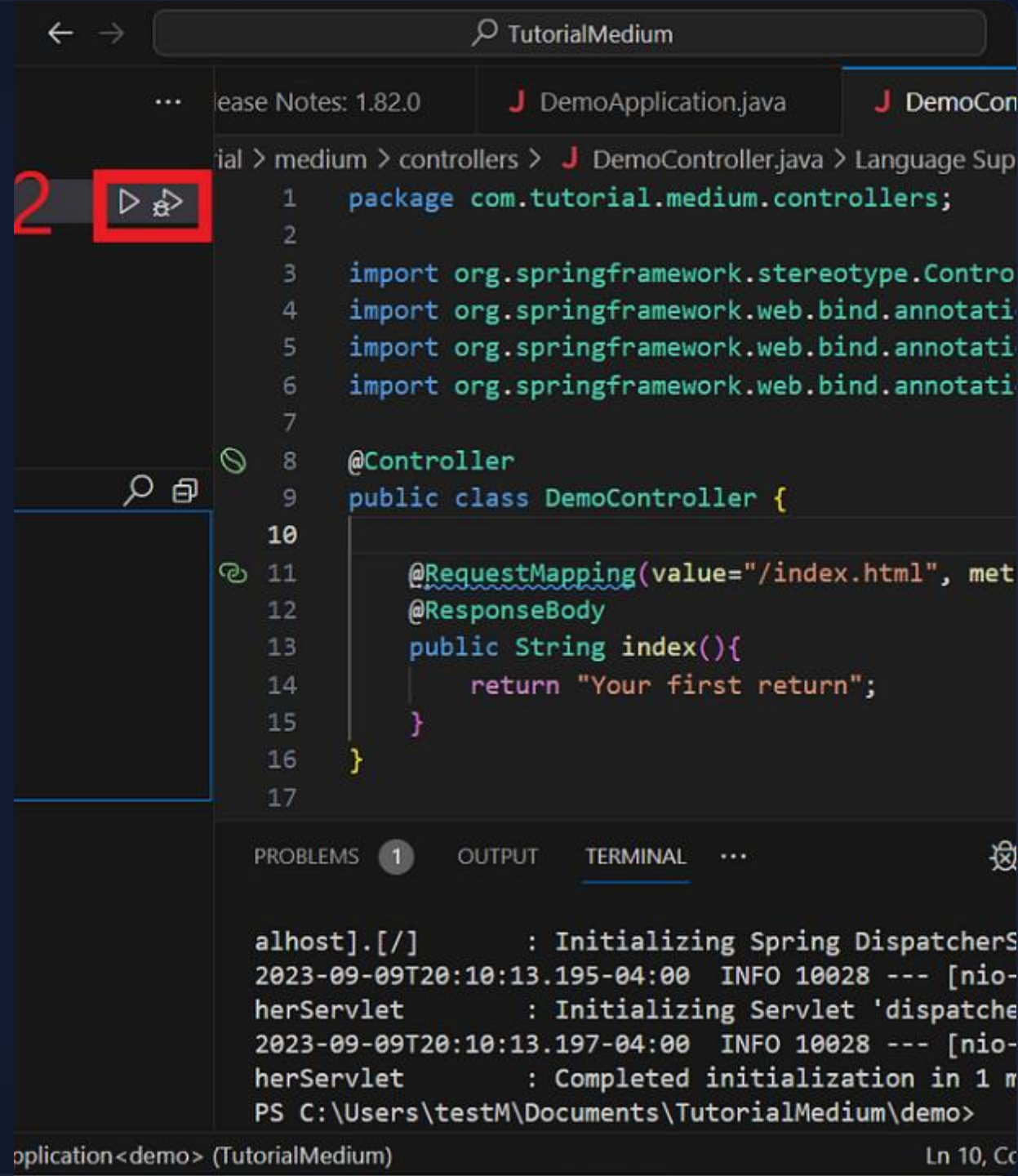
Spring Boot Framework

Rapid Enterprise Development

Unlike legacy Java EE frameworks, Spring Boot doesn't require a large amount of XML configuration.

We used Inversion of Control (IoC) and dependency injection to quickly create production-ready RESTful APIs that are scalable.

This module was all about Controller annotations, Service layers and standalone embedded Tomcat deployment.

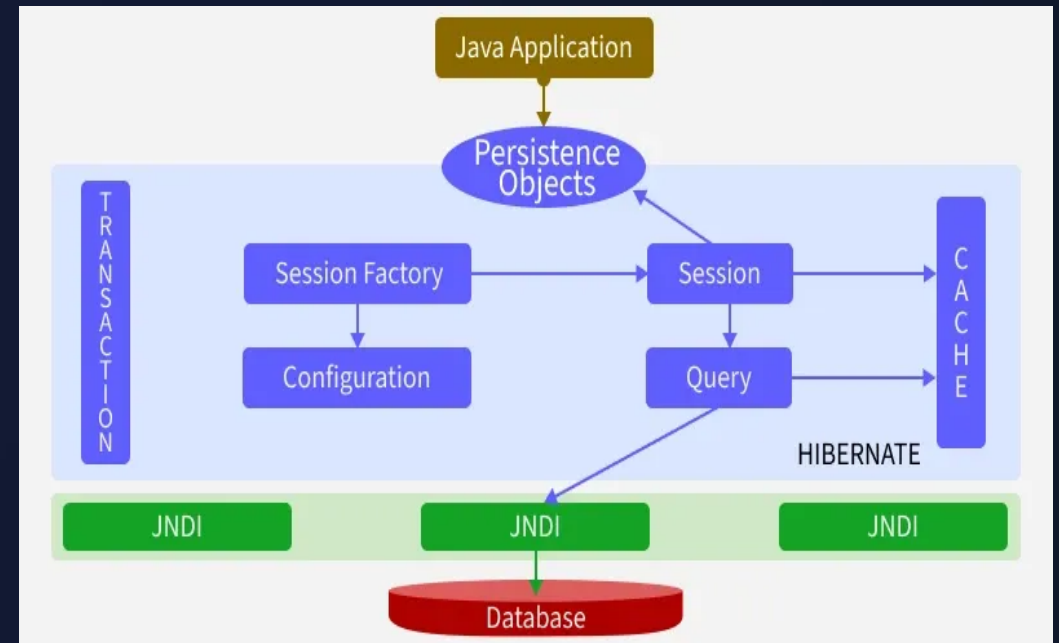


Phase 3: Data Persistence & DevOps

Weeks 7-8: Database Architecture and Version Control

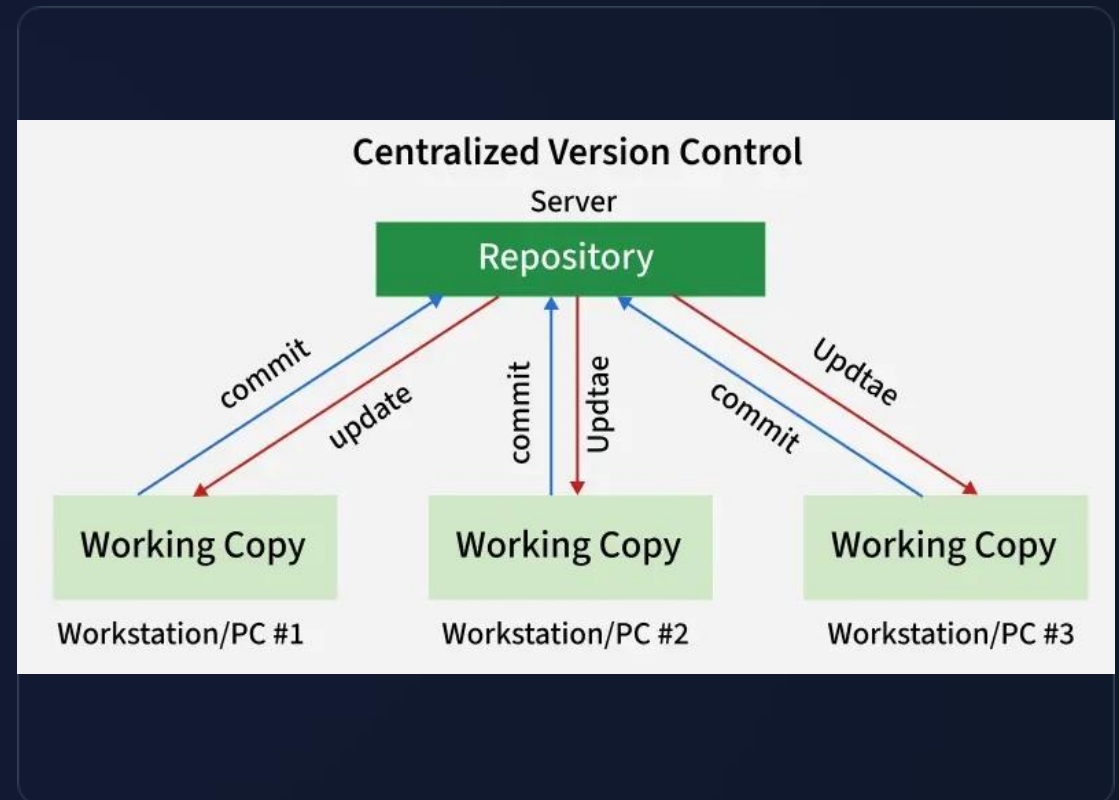
MySQL & Hibernate ORM

- ▶ **Relational Mapping:** Changed the paradigm by allowing to map Java objects (Entities) directly to MySQL data tables by using JPA annotations
- ▶ **Abstraction of SQL:** Removed complex raw SQL strings, and used the Spring Data JPA Repository interfaces to interact with the database.
- ▶ **CRUD Operations:** Designed reliable Create, Read, Update, and Delete operations and guarantee atomic and consistent transactions..
- ▶ **Structured underlying MySQL relational schemas:** No data redundancy using Primary keys and Foreign keys.



Git & Version Control Workflows

- ▶ **Codebase Management:** Standardized software development practices through the tracking of incremental changes to the project source code.
- ▶ **Branching Strategies:** Learned how to isolate new features or bug fixes in separate Git branches so as not to break the main production codebase.
- ▶ **Collaboration:** proficient with standard commands (commit, push, pull, merge) needed for professional collaboration with other developers.
- ▶ **GitHub Deployment:** Released final codebase assignments and documentation to public repositories for evaluation and portfolio building.



Internship Results

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INTENSIVE WEEKS COMPLETED

3

ARCHITECTURAL TIERS MASTERED

Industry Ready Proficiency

The program did a great job at bridging the gap between the academic programming and corporate engineering expectations.

Key outcomes included the ability to independently architect web applications, successfully navigate weekly grade point assessments, and the final achievement of the EduSkills Virtual Internship Certificate.